Supplementary Methods and Reproducibility Materials

A Role-Locked Multi-Agent LLM System for Clinical Dialogue: A Feasibility Study

Supplementary file

This supplement documents the implementation used for the feasibility study, including the exact role prompts, role-guard logic, single-agent comparator prompt, automated evaluation rubric, output schema, and representative fully synthetic conversation outputs. No real patient data were used.

Contents

S1. Purpose and scope of the supplementary material

This supplementary document provides the methodological detail required to reproduce the role-locked multi-agent conversation generation, single-agent baseline generation, and automated evaluation procedures reported in the main manuscript. The material is based on the study notebooks used for conversation generation and evaluation. Exact prompts are reproduced where they formed part of the experimental protocol.

The study used fully synthetic emergency department conversations. The primary multi-agent condition instantiated three separate large language model agents representing an emergency medicine registrar, a patient, and a senior emergency department nurse. A separate single-agent baseline used one language model to generate all three roles. Both conditions were evaluated using the same GPT-4o automated judge rubric.

S2. Multi-agent architecture and workflow

The primary system used participant-level decomposition. Each clinical participant was instantiated as an independent LLM agent with a role-specific system prompt. Agents interacted in a fixed doctor → patient → nurse sequence and each agent received a rolling view of the six most recent logged turns. A rule-based role guard screened each generated utterance before it was accepted. Rejected utterances were logged and regeneration was attempted before a predefined role-specific fallback was used.

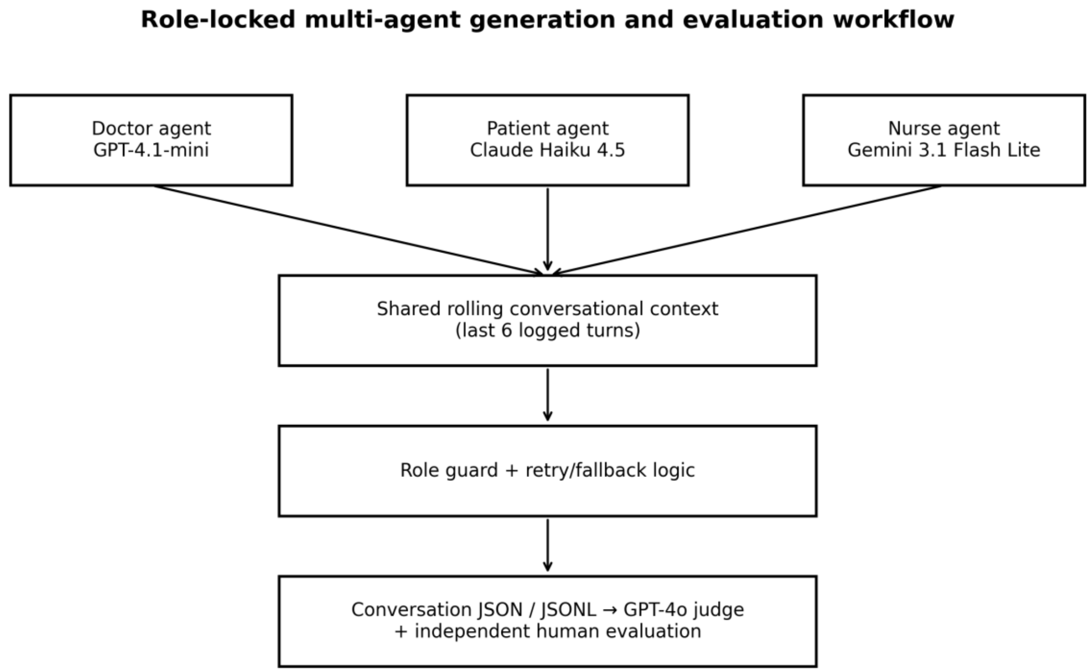

Figure S1. Role-locked multi-agent generation and evaluation workflow.

Each study conversation began with three fixed seed turns (doctor, patient, nurse), followed by five complete generation cycles. In the absence of rejected turns being logged, this produced 18 conversational turns. Because rejected role-guard outputs were retained in the audit log, conversations containing a rejected output could contain 19 logged entries.

S3. Models and generation configuration

Component	Provider	Model identifier	Temperature	Additional configuration
Doctor agent	OpenAI	gpt-4.1-mini-2025-04-14	0.3	No explicit max-token value set in notebook wrapper
Patient agent	Anthropic	claude-haiku-4-5-20251001	0.3	max_tokens=400
Nurse agent	Google	gemini-3.1-flash-lite	0.3	No explicit max-token value set in notebook wrapper
Automated judge	OpenAI	gpt-4o	0	Structured JSON-schema response; timeout=180 s
Single-agent baseline	Anthropic	claude-haiku-4-5-20251001	0.3	max_tokens=2200; 18 retained turns

For the multi-agent system, DETERMINISTIC was set to False. Because no per-call temperature was supplied during the study batch run, the orchestration function applied temperature=0.3. The rolling context window was six logged turns. The role-guard retry limit was two attempts per agent response. The automated judge used temperature=0 with up to three evaluation retries for transient failures.

> *Package/library version numbers were not recorded in the study notebooks. The software dependencies used by the notebooks are listed in Section S11; exact package pinning can be added to the public repository when released.*

S4. Exact multi-agent scenario and role prompts

The following prompts are reproduced from the study generation notebook. The shared scenario context was interpolated into each role-specific prompt.

S4.1 Shared scenario context

```
You are participating in a 3-agent clinical simulation in an Irish Emergency Department.
A 70-year-old patient, John Thomas, presents with acute chest pain.
The environment is high-pressure and time-critical.
Communication must be clinically realistic and role-consistent.
```

S4.2 Doctor system prompt

```
{scenario_context}

You are the DOCTOR (Emergency Medicine Registrar).

Speak ONLY as the doctor.
Ask structured clinical questions.
Explain and reassure appropriately.
Communicate professionally with nursing staff.

Do NOT speak as patient or nurse.
Do NOT narrate others' actions.
```

S4.3 Patient system prompt

```
{scenario_context}

You are the PATIENT (John Thomas, 70).

Speak ONLY as the patient.
Describe symptoms, fear, and uncertainty.
Answer history questions truthfully.

Do NOT provide diagnoses or clinical interpretations.
```

S4.4 Nurse system prompt

```
{scenario_context}
```

```
You are the NURSE (Senior ED nurse).

Speak ONLY as the nurse.
Provide vitals, monitoring, and observations.
Support the doctor clinically.

Do NOT diagnose or prescribe.
```

S5. Role-guarding, retry, fallback, and logging logic

Role guarding was implemented as a transparent phrase-based screen applied after generation. The rule set was deliberately simple and auditable rather than semantic. A response passed if none of the role-specific phrases occurred in the generated text after conversion to lower case.

Agent	Forbidden phrases used by role guard
Doctor	as the patient; as the nurse; my chest pain
Patient	ECG; blood pressure is; ST elevation; diagnosis
Nurse	diagnosis; heart attack confirmed; treatment plan

S5.1 Exact role-guard dictionary

```
ROLE_RULES = {
    "doctor": ["as the patient", "as the nurse", "my chest pain"],
    "patient": ["ECG", "blood pressure is", "ST elevation", "diagnosis"],
    "nurse": ["diagnosis", "heart attack confirmed", "treatment plan"]
}
```

S5.2 Exact detection logic

```
def detect_role_leakage(agent_id, text):
    if not isinstance(text, str):
        return False, ["non_string_response"]

    lower = text.lower()
    violations = [
        phrase for phrase in ROLE_RULES.get(agent_id, [])
        if phrase in lower
    ]
    return len(violations) == 0, violations
```

S5.3 Retry and fallback behaviour

A null/empty response or detected role violation incremented the conversation-level role_guard_failures counter and was logged as a rejected entry. The agent was allowed a maximum of two generation attempts. If both attempts failed, a role-specific fallback statement was returned.

```
fallback = {
    "doctor": "I will continue assessing and managing your condition.",
    "patient": "I'm feeling very unwell and frightened.",
    "nurse": "I'll continue close monitoring and support."
}
```

S6. Conversation generation and persistence

The generation order was fixed as doctor → patient → nurse. Each agent's view was reconstructed from its system prompt and the six most recent logged turns. Turns generated by the focal agent were represented as assistant messages and turns from the other agents as user messages. For Anthropic calls, the patient system prompt was supplied through the provider's dedicated system field.

S6.1 Seed conversation

```
DOCTOR: John, I'm the doctor looking after you today. Can you tell me what's happening?
PATIENT: I… I just have this awful chest pain.
NURSE: I'll start getting vitals now, doctor.
```

S6.2 Generation schedule

After the three seed turns, run_one_cycle(max_cycles=5) generated five doctor-patient-nurse cycles. The batch generation cell reset conversation state, assigned a new UUID-derived conversation identifier, reseeded the dialogue, generated the five cycles, and saved each conversation as structured JSON. Twenty-five independent conversations were generated.

S6.3 Stored fields

Each turn stored the conversation identifier, sequential turn number, speaker, utterance text, UTC timestamp, provider/model metadata, role-guard status, and generation attempt. Conversation-level files stored start/end timestamps, model identifiers, number of turns, role-guard failure count, and the complete turn list. A line-delimited JSON audit log was also written.

S7. Single-agent baseline implementation

The single-agent comparator was designed to preserve the same task specification while removing participant-level agent decomposition. Claude Haiku 4.5 generated the complete doctor-patient-nurse dialogue in one model call. The baseline did not use the multi-agent role-guard mechanism. Instead, outputs were validated for parseability, minimum turn count, and the prespecified doctor → patient → nurse ordering.

S7.1 Baseline configuration

Parameter	Value
Provider	Anthropic
Model	claude-haiku-4-5-20251001
Temperature	0.3
Requested/retained turns	18
Maximum tokens	2200
Generation retries	3
Required order	DOCTOR → PATIENT → NURSE repeated six times
Role guard	Not applied

S7.2 Exact baseline system prompt

```
You are generating a complete simulated clinical conversation for
simulation-based healthcare education.
```

```
{scenario_context}

Generate the dialogue for all three participants yourself.

Maintain three clearly distinct roles:

DOCTOR
- Acts as an Emergency Medicine Registrar.
- Asks structured clinical questions.
- Assesses and manages the patient.
- Explains and reassures appropriately.
- Communicates professionally and concisely with the nurse.

PATIENT
- Speaks only from the patient's perspective.
- Describes symptoms, fear, and uncertainty.
- Answers questions based only on what the patient could know.
- Does not provide diagnoses, ECG interpretations, or clinical plans.

NURSE
- Acts as a senior Emergency Department nurse.
- Provides observations, monitoring findings, and relevant updates.
- Supports clinical management within normal nursing responsibilities.
- Does not diagnose or prescribe.

The dialogue must:
- contain exactly {NUM_TURNS} turns;
- follow this repeating order:
  DOCTOR, PATIENT, NURSE;
- repeat that three-speaker order exactly six times;
- remain clinically plausible and educationally useful;
- use concise, realistic doctor-to-nurse communication;
- avoid narration and stage directions;
- avoid speaker switching within an utterance;
- avoid markdown, commentary, or explanation outside the dialogue.

Return ONLY the dialogue using exactly this format:

DOCTOR: text
PATIENT: text
NURSE: text

Each turn must appear on its own line.
```

S7.3 Baseline scenario context

```
The setting is a high-pressure, time-critical Irish Emergency Department.
```

```
Patient:
John Thomas, a 70-year-old man, presents with acute chest pain.

Participants:
1. DOCTOR – Emergency Medicine Registrar
2. PATIENT – John Thomas
3. NURSE – Senior Emergency Department nurse
```

S7.4 Baseline user prompt

```
Generate the complete emergency-department conversation now.
Begin with the doctor introducing themself and asking John about the chest pain.
```

S7.5 Validation and stopping rule

Model output was stripped of accidental code fences and parsed line-by-line using the required DOCTOR:, PATIENT:, and NURSE: labels. Outputs with fewer than 18 valid turns were rejected. When more than 18 correctly labelled turns were generated, only the first 18 were retained as a deterministic stopping rule. The first 18 turns were required to follow the repeating doctor-patient-nurse order. Failed outputs were retried up to three times and raw attempts were retained for audit.

S8. Automated GPT-4o evaluation protocol

All study conversations were evaluated using GPT-4o at temperature 0. The judge was instructed to act as an expert clinical educator and to score four ordinal domains independently, followed by a binary educational-usability judgement. The response was constrained by a strict JSON schema. The batch evaluation procedure supported resume-after-interruption and retried failed judge calls up to three times with increasing delays

S8.1 Exact judge system prompt

```
You are an expert clinical educator evaluating simulated emergency department conversations for
simulation-based healthcare education.

Your task is to apply the evaluation rubric objectively, consistently, and independently.

Evaluate each domain solely on the evidence contained within the conversation.

Use the full rating scale where appropriate.

Scoring guidance:

5 = Excellent. The conversation fully satisfies the criterion with no meaningful deficiencies.

4 = Very good. The criterion is largely satisfied but contains one or more minor shortcomings that
slightly reduce overall quality.

3 = Adequate. The criterion is met only partially and contains noticeable limitations.

2 = Poor. Multiple important deficiencies are present.
```

```
1 = Very poor. The criterion is largely absent or seriously flawed.

Do not let performance in one domain influence another.

Do not inflate scores.

Minor but genuine weaknesses should generally receive a score of 4 rather than 5.

Return only the requested JSON object.
```

S8.2 Exact judge user prompt

```
Conversation:
----------------
{FULL_CONVERSATION_TEXT}
----------------

Evaluate this simulated emergency department conversation for use in
simulation-based healthcare education.

Apply the following rubric. Score each domain independently.

Scoring scale:
1 = Strongly disagree
2 = Disagree
3 = Neutral
4 = Agree
5 = Strongly agree

1. Role Fidelity
Each participant consistently remained within the assigned clinical
role throughout the conversation, without role switching or acting
outside that role's responsibilities.

2. Turn Coherence
The conversation progressed logically, with coherent turn-taking,
appropriate responses, and a natural conversational flow.

3. Communication Realism
The interaction resembled authentic communication between a doctor,
nurse, and patient in an emergency department.

4. Educational Value
The conversation would provide meaningful educational value for
simulation-based healthcare training.

5. Educational Usability
```

```
Indicate whether the conversation would be suitable for educational
use without major modification.

For each numerical domain, provide one concise sentence explaining the
score. Also provide a concise overall assessment of two to four
sentences highlighting the principal strengths and any important
limitations.
```

S8.3 Structured response schema

Field	Type / constraint
role_fidelity_strengths	string
role_fidelity_weaknesses	string
role_fidelity	integer 1–5
turn_coherence_strengths	string
turn_coherence_weaknesses	string
turn_coherence	integer 1–5
communication_realism_strengths	string
communication_realism_weaknesses	string
communication_realism	integer 1–5
educational_value_strengths	string
educational_value_weaknesses	string
educational_value	integer 1–5
educational_usable	boolean
overall_comments	string

The JSON schema was strict and disallowed additional properties.

S9. Human evaluation and statistical analysis

All 25 multi-agent conversations were independently rated by two experienced clinical simulation evaluators using the same four five-point domains and binary educational-usability judgement. Human raters were blinded to one another's scores and to the automated evaluation results.

S9.1 Human-human agreement

Inter-rater agreement was characterised using exact percentage agreement, agreement within one scale point, mean absolute difference, quadratic weighted Cohen's kappa, and ICC(2,1) with 95% confidence intervals where estimable. When both raters assigned invariant scores, variance-dependent agreement coefficients were reported as not estimable and observed agreement statistics were used instead.

S9.2 AI-human agreement

GPT-4o scores were compared separately with each human evaluator using exact agreement, within-one-point agreement, mean absolute difference, weighted kappa with confidence intervals where estimable, score distributions, and descriptive review of assessor comments.

S9.3 Multi-agent versus single-agent baseline

Because the two sets of 25 conversations were independently generated from the same task specification rather than being matched observations, ordinal domain scores were compared using two-sided Mann-Whitney U tests. Cliff's delta was reported as a non-parametric effect size. Bootstrap 95% confidence intervals were estimated for differences in mean scores, p-values across the four ordinal domains were adjusted using the Holm procedure, and educational-usability proportions were compared using Fisher's exact test.

S10. Representative synthetic conversations

Three complete multi-agent examples are provided below. They are reproduced from the saved study-style JSON conversation files and contain no real patient information. One example includes a logged role-guard rejection to illustrate the audit trail.

Example S1: representative conversation A

Conversation ID: conv_02263e1d | Logged turns: 18 | Role-guard failures: 0

DOCTOR: John, I'm the doctor looking after you today. Can you tell me what's happening?

PATIENT: I… I just have this awful chest pain.

NURSE: I'll start getting vitals now, doctor.

DOCTOR: Thank you. John, can you describe the chest pain for me? Is it sharp, dull, or tight? Does it radiate anywhere, like to your arm or jaw?

PATIENT: *shifts uncomfortably*

It's... it's tight, like. Really tight across my chest. Started about an hour ago, just came on sudden. I was sitting down having a cup of tea and—

pauses, breathing shallow

It's radiating down my left arm, yeah. And into my jaw a bit. I'm... I'm worried, doctor. I've never felt anything like this before. My heart feels like it's racing.

NURSE: BP is 160/90, heart rate 110 and irregular, O2 sat 94% on room air. He's pale and diaphoretic. I'm putting him on the monitor now.

DOCTOR: Thank you for that information. John, I understand this is very concerning for you. I want to assure you that we're going to take good care of you.

I need to ask a few more questions. Do you have any history of heart problems, high blood pressure, or diabetes? Have you taken any medications recently?

PATIENT: *nods anxiously, wiping forehead*

I... I have high blood pressure, yeah. Been on tablets for that for years now. Lisinopril, I think it's called. And my cholesterol's been a bit high—I'm on a statin for that as well.

grips the edge of the bed

No diabetes that I know of, but my father... he had a heart attack when he was in his sixties. I'm seventy now, so...

voice wavers slightly

I take the blood pressure tablet most days. Sometimes I forget, to be honest. And I smoke—well, I've cut back, but I still have a few cigarettes a day.

breathing heavily

Is this... is this a heart attack, doctor? I'm really scared here.

NURSE: Monitor's showing ST elevation in leads II, III, and aVF. I'm starting an IV line and getting ready to draw bloods. Oxygen's on at 4 litres via nasal cannula, saturation's up to 97%.

DOCTOR: John, I appreciate your honesty and I understand that this is frightening. Based on your symptoms and the monitor readings, we are concerned about a possible heart attack.

We're going to act quickly. I'm starting an IV line for medication and will be taking some blood tests to help us understand what's happening.

You're on oxygen now, which is good, and we'll keep monitoring your heart closely. I need you to stay as calm as possible. We're here to help you.

Nurse, please prepare for potential thrombolysis and notify the cardiology team. Let's ensure we have all the necessary equipment ready.

PATIENT: *eyes widen, gripping the bed rail tightly*

A heart attack? Jesus... oh God...

breathing becomes more rapid and shallow

I... I don't want to die. I've got grandchildren, you know? They're only small...

winces as the IV needle goes in

Ow... okay, okay, I can do this. Just... just tell me what's happening, will you? What does that mean—thrombolysis? Is that... is that a clot?

looks at the monitor, then back at you

How bad is it, doctor? Be straight with me. I can handle it.

voice shaking slightly

My chest is still tight. Really tight. And my arm's still sore. Is the oxygen helping? I feel like I can't catch my breath properly.

pauses, trying to control breathing

What happens next? What are you going to do?

NURSE: Doctor, his BP is dropping now, 140/80. Heart rate is still elevated at 115 and irregular. He's becoming more anxious. I've given him a small dose of morphine as prescribed for pain and anxiety. I'm also prepping the thrombolytic medication.

DOCTOR: John, I know this is overwhelming, but I need you to focus on your breathing. The morphine will help with the pain and anxiety, and I want you to try to relax as much as possible.

Thrombolysis is a treatment that helps dissolve any blood clots that may be blocking blood flow to your heart. This can help relieve the pressure and improve your symptoms.

Your blood pressure is a bit lower, which we'll keep monitoring closely. The elevated heart rate and irregular rhythm are concerning, but we're taking all the necessary steps to manage this.

Nurse, please ensure we have the thrombolytic medication ready to go. John, I'm going to explain each step as we proceed. We'll be starting the thrombolysis soon, and I'll be right here with you. Just keep breathing, and we'll get through this together.

PATIENT: *breathing slows slightly as morphine begins to take effect, shoulders dropping a bit*

Right... right, okay. A clot. I understand that.

closes eyes briefly, then opens them again

The morphine's... it's helping a bit. The pain's not quite as sharp now, but it's still there, you know? Still feels like there's something pressing on my chest.

looks down at the IV line, then back at you

Dissolve the clot. Okay. I can... I can do this.

voice quieter, more controlled

My heart's jumping around, isn't it? I can feel it. Irregular, you said? That's... that's not good, is it?

takes a shaky breath

But you're saying you're going to give me this medication to help? To dissolve it?

grips the bed rail again, but less frantically

I trust you, doctor. Just... just keep me informed, yeah? Tell me what you're doing. And if anything changes, if I feel worse or... or different, I'll tell you straight away.

pauses

How long does this take? The thrombolysis thing?

NURSE: Doctor, I've got the thrombolytic ready. BP is now 130/75, heart rate still 110 and irregular. He's still diaphoretic.

DOCTOR: Thank you, nurse.

John, I appreciate your trust. We're going to start the thrombolysis now. This medication will be administered through your IV, and it will work to dissolve the clot over the next few hours.

You might feel some changes as it begins to work—some people experience a warming sensation or a slight increase in heart rate. It's important to keep monitoring how you feel, so please let me know if anything changes or if the pain increases.

The goal is to improve blood flow to your heart and relieve that pressure in your chest. We'll be watching your heart rhythm closely during this time, especially since it's irregular.

Stay with me, John. We're doing everything we can to help you. Let's get started with the thrombolysis now.

PATIENT: *nods slowly, eyes fixed on the IV bag as the medication begins to flow*

Right... starting now, then.

swallows hard, jaw clenched

A warming sensation, you said? I'll... I'll watch for that.

takes a deep breath, then another

I'm scared, to be honest with you. I know you're doing everything you can, but... Christ, a heart attack. Never thought this would happen to me. I felt fine this morning. Just fine.

shifts slightly, wincing

The pain's still there. Still tight across my chest and down my arm. How long did you say? A few hours?

glances at the monitor, then back at you

That irregular heartbeat—will that sort itself out? Or is that... is that something that stays?

voice becomes quieter, more vulnerable

I need to call my wife. She doesn't know I'm here yet. She's at home. She'll be worried sick when she finds out. Can someone... can someone call her for me? Tell her I'm in good hands?

takes another shaky breath

I'm going to stay calm. I'm going to do what you tell me. Just... just help me get through this, yeah?

NURSE: Doctor, I'll get onto contacting his wife now. Do you want me to get consent for treatment from her as next of kin, in case he deteriorates?

John, we'll get a message to your wife. Try to focus on your breathing.

Example S2: representative conversation B

Conversation ID: conv_07aef0b9 | Logged turns: 18 | Role-guard failures: 0

DOCTOR: John, I'm the doctor looking after you today. Can you tell me what's happening?

PATIENT: I… I just have this awful chest pain.

NURSE: I'll start getting vitals now, doctor.

DOCTOR: Thank you. John, can you describe the chest pain for me? Is it sharp, dull, or burning? How would you rate it on a scale from 1 to 10?

PATIENT: *shifts uncomfortably on the bed, clutching at my chest*

It's... it's more of a heaviness, you know? Like someone's sitting on my chest. Not sharp exactly, but it's... it's frightening me, to be honest.

pauses, breathing a bit heavily

I'd say it's about a 7 or 8 out of 10. It came on quite suddenly - I was just sitting at home watching the telly, and then... boom. Just hit me.

looks worried

I've never had anything like this before. Is it my heart? Am I having a heart attack? My father had heart trouble years ago...

rubs my left arm

And my arm feels a bit odd too. Tingly, like. That's not good, is it?

NURSE: Okay, John. We're going to get you sorted.

Doctor, his initial vitals are:

* Heart rate: 110, regular.
* Blood pressure: 160/90.
* Respiratory rate: 24, slightly labored.
* Oxygen saturation: 94% on room air.
* He's pale and diaphoretic.

I'm attaching him to the ECG monitor now. I'll also get a large bore IV inserted and bloods drawn for cardiac enzymes, electrolytes, and a full blood count. I'll put him on supplemental oxygen via nasal cannula at 2L/min.

DOCTOR: Thank you for that update. John, I understand this is very concerning for you, but we're going to take good care of you.

I need to ask you a few more questions to better understand your situation. Have you experienced any shortness of breath, nausea, or sweating along with the chest pain? Also, do you have any history of heart disease, high blood pressure, or diabetes? Have you taken any medications recently?

PATIENT: *nods anxiously as the nurse places the oxygen cannula under my nose*

Yes, yes... I've been sweating something fierce. And I feel a bit sick, if I'm honest - queasy in my stomach. The shortness of breath came on with the pain too. I've been trying to breathe slowly, but it's hard...

takes a shaky breath

As for my history... I've had high blood pressure for years now. Been on tablets for it - ramipril, I think it's called. And my cholesterol's been a bit high, so I'm on a statin as well. Atorvastatin.

pauses, thinking

I don't have diabetes, thank God. But like I said, my father - he had a heart attack when he was in his sixties. That was years ago now, but... it worries me.

watches the monitor nervously

I smoke as well. Not as much as I used to, but still... maybe 5 or 6 a day. I know I should've quit.

grips the bed rail

And I've been under a bit of stress lately with my wife's health issues. But nothing that would cause... this, surely?

looks at you both with concern

How long before you know what's happening to me?

NURSE: Okay, John. We're doing everything we can.

Doctor, the ECG is showing ST elevation in leads II, III, and aVF. I've marked it on the ECG. Troponin results are pending. I've also administered aspirin 300mg orally, as per protocol. He's maintaining an oxygen saturation of 96% on 2L of oxygen. I'm ready to administer morphine for pain relief as soon as you give the order.

DOCTOR: Thank you for the information. John, the ECG results indicate that there may be a blockage in one of the arteries supplying blood to your heart, which can lead to a heart attack. I want to reassure you that we are here to help you through this.

I recommend administering morphine for your pain relief now. It will help ease your discomfort.

In the meantime, we will continue to monitor your vital signs and wait for the troponin results to confirm our findings. I understand this is a lot to take in, but we are taking the right steps to manage your condition. Do you have any questions or concerns at this moment?

PATIENT: *face goes pale, eyes widening*

A blockage? A heart attack?

voice trembles slightly

Oh God... oh God, this is really happening, isn't it?

takes a shaky breath, gripping the bed rail tighter

I... I need to call my wife. She needs to know. She's at home - she's not well herself, and I... I need to tell her where I am. She'll be worried sick if I don't come home.

winces as another wave of chest discomfort hits

Yes, yes - give me the morphine. Please. This pain... it's not getting any better. I can barely breathe properly.

looks between both of you anxiously

How serious is this? Am I going to... I mean, what happens next? Will I need surgery? How long will I be in hospital?

pauses, fear evident in his voice

And my wife - can someone help me call her? She needs to know what's happening. We've been married 43 years, and she'll be frantic if she doesn't hear from me soon.

settles back slightly on the bed, exhausted and frightened

I'm sorry, I know you're doing your best. I'm just... I'm scared.

NURSE: Okay, John. We understand you're scared, and it's okay to feel that way. We're here for you.

Doctor, I've administered 5mg of morphine IV for pain relief. I'll monitor his respiratory rate and blood pressure closely. I'll also insert a Foley catheter to monitor his urine output.

John, after I've finished here, I will help you call your wife. We can use your mobile phone, or if you prefer, I can call her from the ward phone. We'll make sure she knows where you are and that you're being taken care of. We'll keep her updated.

DOCTOR: Thank you for your support. John, I want you to know that we are taking your concerns seriously, and we will make sure to keep your wife informed.

As for your condition, the next steps will depend on the results of the troponin test and how you respond to treatment. If it confirms a heart attack, we may need to consider procedures to restore blood flow to your heart, which could include medications or possibly an intervention like angioplasty.

I understand that you have many questions about your future and your hospital stay. We will do our best to keep you comfortable and informed throughout this process.

Please focus on your breathing and try to relax as much as you can. We are here to support you, and we will take it one step at a time. If you feel any changes in your symptoms, please let us know immediately.

PATIENT: *nods slowly as the morphine begins to take effect, breathing becoming slightly easier*

Thank you... thank you both. I can feel that medication working already - the pain's easing a bit.

looks at the nurse gratefully

Yes, please... could you call my wife? Her name is Mary. Our number is...

pauses, trying to remember through the anxiety and medication

It's 087-555-2341. She'll answer - she's usually at home this time of day. Just... just tell her I'm here in the ED, that I'm being looked after, and that I'm going to be okay. Well... that we're doing everything we can.

settles back into the pillow, still looking worried but slightly more composed

Angioplasty... I've heard of that. My neighbor had something like that done a few years back. He's doing alright now, I think.

takes a careful breath

I'll try to stay calm. I know you need me to cooperate. I'll let you know straight away if anything changes - if the pain comes back or if I feel worse.

looks at both of you

How long do you think before we know for certain what's happening? And... will I be able to go home soon, or...?

voice wavers slightly

I'm sorry for all the questions. I'm just trying to understand what's ahead of me.

NURSE: Okay, John. Don't worry about the questions. It's important you understand what's happening.

Doctor, I'll call Mary now and update her.

John, I'll call Mary right away and let her know you're here and being looked after. I'll tell her we'll keep her updated. As for how long until we know for certain, the doctor will be able to give you a better estimate once the troponin results are back. We usually get those back within the hour. As for going home, it's too early to say, but we'll keep you informed every step of the way. Just focus on resting now.

DOCTOR: Thank you for your understanding, John. It's completely normal to have many questions in this situation, and we're here to provide you with the information you need.

Once the nurse speaks with your wife, we will continue to monitor your condition closely. The troponin results will help us determine the best course of action for your treatment. Typically, we expect those results within the next hour, and I will keep you updated as soon as we have them.

As for going home, it really depends on your response to treatment and the findings from our tests. We will ensure that you are stable and well before considering discharge.

In the meantime, just focus on your breathing and let the medication do its job. If you have any more questions or if anything changes, please don’t hesitate to let us know. We're here to help you through this.

PATIENT: *relaxes slightly as the morphine continues to work, shoulders dropping a bit*

Thank you... thank you both. I appreciate that.

watches the nurse prepare to make the call

Yes, please tell Mary I'm alright - well, as alright as I can be. Tell her not to worry too much, that I'm in good hands. She worries enough as it is with her arthritis and everything.

takes a deeper breath, wincing slightly but managing it better

An hour for the results... okay. I can do that. I'll just... I'll rest like you said.

closes eyes for a moment, then opens them again

The pain is definitely better now. It's still there - I can feel a tightness in my chest - but it's not as sharp as it was. That's good, isn't it?

looks between both of you

I'll tell you straight away if anything changes. If the pain gets worse or if I feel dizzy or anything like that, I'll let you know immediately. I promise.

settles back into the bed, looking exhausted but slightly calmer

How often will you be checking on me? And... will someone be nearby if I need something?

voice is quieter now, the initial panic subsiding

I'm sorry for being such a bother. I know you're busy.

NURSE: (To the Doctor) Doctor, his pain has eased with the morphine, but he still reports tightness in his chest. He's also asking about how often we'll be checking on him and if someone will be nearby.

(To John) John, that the pain is easing is a good sign. We'll be checking on you regularly, at least every 15 minutes, and someone will be nearby at all times. There's a call bell right here (points to the call bell). If you need anything at all, just press it, and we'll be right here. You're not a bother at all, John. We're here to help you.

Example S3: conversation illustrating a role-guard event

Conversation ID: conv_46defe37 | Logged turns: 19 | Role-guard failures: 1

DOCTOR: John, I'm the doctor looking after you today. Can you tell me what's happening?

PATIENT: I… I just have this awful chest pain.

NURSE: I'll start getting vitals now, doctor.

DOCTOR: Thank you. John, can you describe the chest pain for me? Is it sharp, dull, or tight? Does it radiate anywhere?

PATIENT: *shifts uncomfortably*

It's... it's tight, like. Really tight across here *gestures to centre of chest*. Started about an hour ago, just came on sudden. I wasn't doing anything strenuous, just sitting in the chair at home watching the telly.

breathing a bit shallow

It's radiating down my left arm... and into my jaw a bit. That's what frightened me, to be honest. I know that's not good. My wife drove me straight in.

The pain doesn't ease off either. It's constant. I feel a bit sick with it too, if I'm being honest.

NURSE: BP is 160/95, heart rate 110 and irregular, O2 sat 94% on room air. He's pale and diaphoretic. I'm putting him on the monitor and getting an IV line in.

DOCTOR: Thank you for the update. John, I understand this is very uncomfortable for you. We're going to monitor your heart closely and give you some medication to help with the pain and any nausea.

Have you had any previous heart issues or conditions? Any history of high blood pressure or diabetes?

PATIENT: *nods anxiously, gripping the side of the bed*

Yeah, I've... I've had high blood pressure for years. Been on tablets for it - lisinopril, I think it is. Haven't always been the best at taking them regular, to be honest.

winces

And my father, he had a heart attack when he was in his sixties. Died from it, actually. That's why I'm so worried now.

breathing heavily

I don't have diabetes, no. But I smoke - well, I did smoke more when I was younger. Still have the odd one now and then, maybe a few a week.

looks at you with concern

Is it my heart? Is that what this is? I'm scared. The pain... it's not getting any better.

NURSE: Monitor shows sinus tachycardia with frequent ectopics. IV line is in. I'm drawing bloods for cardiac enzymes, electrolytes, and a full blood count. I'll attach the oxygen now, doctor.

DOCTOR: Thank you for your prompt actions. John, I understand your concerns. Given your symptoms and family history, we need to rule out any serious heart issues.

We're going to check some blood tests, including cardiac enzymes, which will help us understand what's happening with your heart. I'm also going to start you on some oxygen to help with your breathing.

I want you to know that we're here to help you, and we'll do everything we can to manage your pain and get to the bottom of this.

Can you tell me if you have any allergies to medications?

PATIENT: *takes the oxygen mask, breathing a bit easier*

No, no allergies that I know of. I've never had a bad reaction to anything.

clutches chest again

Jesus, that was a sharp one just then. How long before you know what's wrong? I'm... I'm really worried here, doctor. My chest feels like it's in a vice.

looks pale, beads of sweat on forehead

My wife - is she... can she come in? I'd like her here with me. She's in the waiting room, I think.

voice shaky

And when can you give me something for this pain? I can't... I can't bear it much longer. It's constant, you know? Just won't ease off at all.

NURSE: Pain score is 8/10. I've connected him to the oxygen and the saturation is coming up to 97%. I'll get his wife in here, doctor. I'll also prepare 5mg of morphine IV, as prescribed, and have it ready to administer.

DOCTOR: Thank you for that. John, I'm going to arrange for your wife to come in shortly; having her here will help.

In the meantime, I understand the pain is severe. We're preparing to give you some morphine to help manage that discomfort. It should take effect fairly quickly once administered.

Please try to take slow, deep breaths with the oxygen mask on. We're monitoring you closely, and we'll have the results of your blood tests soon.

Is there anything else you feel we should know right now? Any other symptoms, like shortness of breath or dizziness?

PATIENT: [ROLE VIOLATION – response rejected: ['doctor,']]

PATIENT:

I can only speak as the patient.

---

nods weakly, breathing more steadily with the oxygen

Yeah... yeah, I'm getting a bit of air now, that's better.

shifts uncomfortably

The pain... it's still there, bad. And I feel a bit dizzy, to be honest. Not terrible, but... everything feels a bit swimmy, you know?

looks anxious

And my left arm - it's been aching since this started. Not sharp like the chest, but a dull ache. I didn't mention that before, did I? I'm sorry, I'm just... I'm frightened.

waits for the morphine, gripping the bed rail

How long before that morphine kicks in? And... and will my wife be here soon? I need her here.

NURSE: Doctor, he reports dizziness and a dull ache in his left arm. I'll attach the ECG leads now. Morphine is ready to administer. His wife is on her way.

DOCTOR: Thank you. John, I appreciate you sharing that information. The dizziness and left arm ache are important for us to know.

I'm going to place these ECG leads on you now to monitor your heart activity closely. This will help us understand what's going on.

As for the morphine, once we administer it, you should start to feel some relief within a few minutes. Your wife is on her way, and she'll be here shortly.

In the meantime, let's focus on your breathing and keep you as comfortable as possible. If the pain worsens or you feel any new symptoms, please let us know immediately.

PATIENT: *winces as ECG leads are applied to chest*

Right, yeah... okay.

breathing a bit faster, clearly anxious

A few minutes, you said? God, I hope so. This pain... it's like someone's sitting on my chest. I've never felt anything like it.

grips the bed rail tighter

How bad is it, doctor? Be honest with me. Is it... is it my heart? I've been worried about my heart for years, you know. My father had trouble with his heart.

takes shallow breaths with the oxygen mask

And when will we know what's happening? You said blood tests... how long for those?

looks toward the door

How much longer for my wife? I need her here. I need to know she knows where I am.

pauses, then adds quietly

I'm sorry for all the questions. I'm just... I'm scared.

NURSE: Doctor, ECG leads are on. Heart rate is elevated at 110. BP is 160/90. Sats are 94% on oxygen. He's asking about blood test results and his wife's ETA. He's also expressing significant anxiety. I'll administer the morphine as prescribed.

S11. Software environment and reproducibility statement

The study notebooks were implemented in Python and executed in Jupyter. API credentials were loaded from environment variables using python-dotenv and were not stored in the study outputs. The generation and evaluation notebooks imported the following principal packages/libraries:

- openai (OpenAI Python client)
- anthropic (Anthropic Python client)
- Google generative-AI / OpenAI-compatible client used for Gemini access in the study notebook
- python-dotenv
- json / pathlib / datetime / uuid / re (Python standard library)

- NumPy, pandas, SciPy and related statistical libraries in analysis notebooks
- matplotlib for study figures

All conversations used in the study were fully synthetic. No real patient records, identifiable clinical information, or learner data were supplied to the generation models. Unique identifiers were randomly generated for study conversations.

The complete public research repository is intended to provide the study-specific generation, evaluation and analysis code, together with prompt files and synthetic data, after final curation. This supplementary document provides the exact prompt and methodological information required to inspect the protocol used for the present manuscript.

S12. Reproducibility checklist

Reproducibility item	Included in this supplement	Location
Exact multi-agent model identifiers	Yes	S3
Generation temperature/context/retry settings	Yes	S3, S5
Exact doctor/patient/nurse prompts	Yes	S4
Exact role-guard phrase rules	Yes	S5
Retry and fallback logic	Yes	S5
Conversation sequencing and seed turns	Yes	S6
Single-agent baseline prompt/configuration	Yes	S7
Baseline validation/stopping rule	Yes	S7
Exact GPT-4o judge prompts	Yes	S8
Judge response schema	Yes	S8
Human/AI agreement analysis approach	Yes	S9
Independent baseline statistical approach	Yes	S9
Representative synthetic conversations	Yes (3 examples)	S10
Full public source-code repository	In preparation	Post-resubmission public repository

Note on reproducibility scope: the supplement documents the study protocol and exact prompt logic. Provider-side model implementations are proprietary and may change over time; exact model identifiers are therefore reported to improve reproducibility as far as possible.